

SANCTI SPIRITUS, Cuba

WMO: 783490

Lat: 21.933N

Lon: 79.450W

Elev: 97

StdP: 100.17

Time Zone: -5.00 (CUB)

Period: 86-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	14.6	9.3	7.4	16.8	10.8	8.2	18.3	9.3	22.6	8.3	23.4	1.1	320	0.245

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.7	33.1	24.9	32.5	24.7	31.9	24.6	26.5	30.6	26.2	30.4	25.8	30.1	2.0	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.7	28.4	25.0	20.3	28.2	24.7	19.9	27.8	83.0	30.8	81.4	30.5	80.2	30.2	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.2	5.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.2	22.4	23.1	23.3	25.4	26.2	26.9	27.4	27.6	26.9	25.8	24.2	22.6
	DBStd	2.51	2.26	2.26	2.31	1.51	1.18	1.16	0.95	0.91	0.86	1.32	1.82	2.37
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	9	3	2	1	0	0	0	0	0	0	0	0	3
	CDD10.0	5533	384	366	412	461	501	508	541	547	508	489	427	391
	CDD18.3	2501	129	134	154	211	242	258	282	288	258	231	177	136
	CDH23.3	19110	721	878	1109	1663	1901	2168	2547	2656	2075	1619	1075	699
	CDH26.7	6323	144	224	318	586	684	759	991	1070	712	463	248	123
Wind	WSAvg	1.6	1.9	1.9	2.1	2.0	1.6	1.3	1.3	1.3	1.2	1.4	1.7	1.8
Precipitation	PrecAvg	1492	32	47	46	56	236	257	160	201	210	185	54	25
	PrecMax	1985	122	189	239	228	622	503	308	662	335	540	173	118
	PrecMin	1082	0	1	2	1	71	82	47	0	83	38	0	0
	PrecStd	231	30	51	44	47	142	113	72	108	64	112	46	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	31.1	32.0	32.8	33.2	33.1	33.8	34.3	32.7	31.9	30.9	30.1
		MCWB	21.9	21.6	21.9	22.2	23.4	25.6	25.4	25.1	25.1	25.3	24.0	22.6
	2%	DB	29.2	30.1	30.7	31.7	32.1	32.2	32.8	33.1	31.9	31.0	29.9	29.0
		MCWB	21.7	21.4	21.7	22.2	23.0	25.3	25.2	25.4	25.3	25.1	24.0	22.5
	5%	DB	28.4	29.2	29.7	31.0	31.3	31.4	32.1	32.3	31.2	30.2	29.1	28.1
		MCWB	21.7	21.4	21.6	22.3	23.0	25.1	25.3	25.3	25.3	24.9	23.8	22.2
	10%	DB	27.3	28.2	28.6	30.0	30.3	30.6	31.2	31.5	30.5	29.4	28.2	27.0
		MCWB	21.3	21.3	21.4	22.0	23.0	24.9	25.2	25.3	25.3	24.8	23.5	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.6	23.4	24.4	25.2	25.7	26.6	26.7	27.0	26.7	26.3	25.5	24.1
		MCDB	27.6	27.6	28.0	28.4	30.6	30.6	31.0	31.2	30.3	29.9	29.1	27.4
	2%	WB	23.0	22.7	23.4	24.1	25.0	26.1	26.2	26.5	26.2	25.8	24.9	23.5
		MCDB	27.0	27.1	27.7	27.9	29.1	30.3	30.7	30.8	30.1	29.5	28.2	26.9
	5%	WB	22.5	22.4	22.7	23.5	24.4	25.7	25.8	26.1	25.9	25.5	24.4	23.1
		MCDB	26.2	26.7	26.9	27.7	28.4	29.8	30.3	30.5	29.9	29.0	27.8	26.4
	10%	WB	22.0	22.0	22.3	23.0	24.0	25.3	25.5	25.7	25.5	25.1	24.0	22.6
		MCDB	25.5	26.1	26.5	27.6	27.8	29.2	29.8	30.0	29.3	28.5	27.2	25.7
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	10.4	9.8	8.8	7.7	8.5	8.7	8.0	7.4	7.9	8.7
		MCDBR	10.3	11.0	11.3	10.6	10.3	8.6	9.4	9.6	8.7	8.5	8.7	9.3
		MCWBR	3.7	3.5	3.6	2.8	2.9	3.0	3.2	3.3	3.2	3.4	3.5	3.6
	5% WB	MCDBR	9.2	9.8	9.5	8.9	8.3	7.9	8.6	8.8	8.2	7.6	7.7	8.5
		MCWBR	3.8	3.6	3.7	3.1	3.1	3.2	3.4	3.6	3.5	3.4	3.6	3.7
Clear-Sky Solar Irradiance	taub	0.359	0.366	0.382	0.403	0.425	0.459	0.471	0.442	0.417	0.402	0.381	0.377	
	taud	2.523	2.503	2.446	2.381	2.367	2.300	2.246	2.354	2.433	2.466	2.477	2.488	
	Ebn at Noon	912	927	922	901	872	836	826	854	876	878	880	878	
	Edh at Noon	96	104	115	124	125	133	140	126	115	107	100	97	
All-Sky Solar Radiation	RadAvg	4.21	5.11	5.93	6.50	6.25	5.99	6.51	6.27	5.60	4.82	4.32	3.94	
	RadStd	0.23	0.19	0.22	0.19	0.60	0.40	0.34	0.22	0.29	0.34	0.25	0.29	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page